BRITISH NORTH SEA GAS & LNG-LINKED DERIVATIVES

An investment strategy prospectus framework for pricing future gas output, managing downside price risk and capturing value from a potential medium-term demand recovery.

01

COST-DOWN THESIS

Lower unit costs may widen project margins - unless benchmark prices fall faster.

02

DEMAND OPTIONALITY

Growth in power, industry and LNG trade can offset supply additions.

03

DERIVATIVES FIRST

Launch on established NBP contracts; make any new LNG index conditional on physical capacity.

PHYSICAL-MARKET SCOPE NOTE

The UK currently produces North Sea natural gas, not LNG domestically. “New North Sea LNG” is treated here as a conditional future export/infrastructure concept and/or LNG-linked financial product. NBP gas derivatives can be pursued first; physical LNG sales require liquefaction, shipping and export arrangements.

INVESTMENT OVERVIEW

Investment case in one page

A staged derivatives proposition: monetise price uncertainty now, while preserving an option on future North Sea LNG infrastructure.

RECOMMENDATION

Build an NBP-centred gas risk-management programme for North Sea producers and industrial buyers. Add NBP-TTF and TTF-JKM spread products as client demand develops. Treat a bespoke "North Sea LNG" derivative as a later-stage product, subject to physical feasibility, index governance and regulatory review.

43%
Domestic gas share
of UK gross gas supply in 2024

14%
LNG import share
of UK gross gas supply in 2024

~6%
Jackdaw peak uplift
expected increase in domestic production

-0.5%
2026 demand outlook
IEA forecast for global gas demand

INVESTMENT THESIS

- Cost advantage: selected new developments and operating efficiencies could lower the incremental cost of marketable North Sea gas; this must be proven project by project.
- Demand upside: a medium-term recovery in industrial activity, power demand and LNG trade could absorb new supply, supporting realised prices.
- Commercial monetisation: forward sales, options and basis hedges sell certainty to producers and consumers, rather than depend on Barclays taking a directional bet.

DECISION DISCIPLINE

- Near-term caution: the IEA expects global gas demand to fall by about 0.5% in 2026 amid high prices and supply disruption.
- Physical constraint: there is no current UK domestic LNG production chain; export LNG needs liquefaction and shipping infrastructure.
- Underwrite in stages: launch standard NBP exposures first. Do not market physically deliverable "North Sea LNG" until capacity and delivery terms exist.

INVESTMENT VIEW | The opportunity is credible as a risk-intermediation business. A British LNG export proposition is a separate infrastructure investment and should remain conditional.

MARKET STRUCTURE

Start with the physical commodity

The first product must match what the UK actually produces and where it can be delivered.

01 UPSTREAM

North Sea gas

Field production, gathering, processing and pipeline entry.

→

02 UK GAS MARKET

NBP delivery

Existing benchmark for British wholesale gas; immediately relevant to producers and buyers.

→

03 FUTURE LNG

Liquefaction + export

Requires new capacity, export economics, shipping access and counterparties.

MARKET FACTS RELEVANT TO THE PROSPECTUS

Fact	What it means for the strategy
UK LNG status	The UK Government stated in April 2026 that the UK produces natural gas but does not produce LNG domestically; LNG imports are regasified into the gas system. [1]
2024 supply mix	Domestic production represented 43% of gross UK gas supply; LNG imports accounted for 14%, with most of the remainder coming by pipeline, principally from Norway. [1]
Jackdaw approval	On 21 October 2026, the Government authorised the regulator to consent to production; official statements say output could begin this winter and add around 6% to domestic production at peak. [2]
Benchmark availability	ICE lists NBP futures for physical delivery at the UK virtual trading point, with monthly, quarterly, seasonal and calendar strips. [3]

IMPLICATION

The near-term investable product is not a claim on a UK LNG cargo. It is a contract whose payoff references an established gas price or spread. A new physical North Sea LNG chain would require a separate feasibility case covering liquefaction cost, energy use, permits, carbon intensity, capex, capacity access, shipping and offtake.

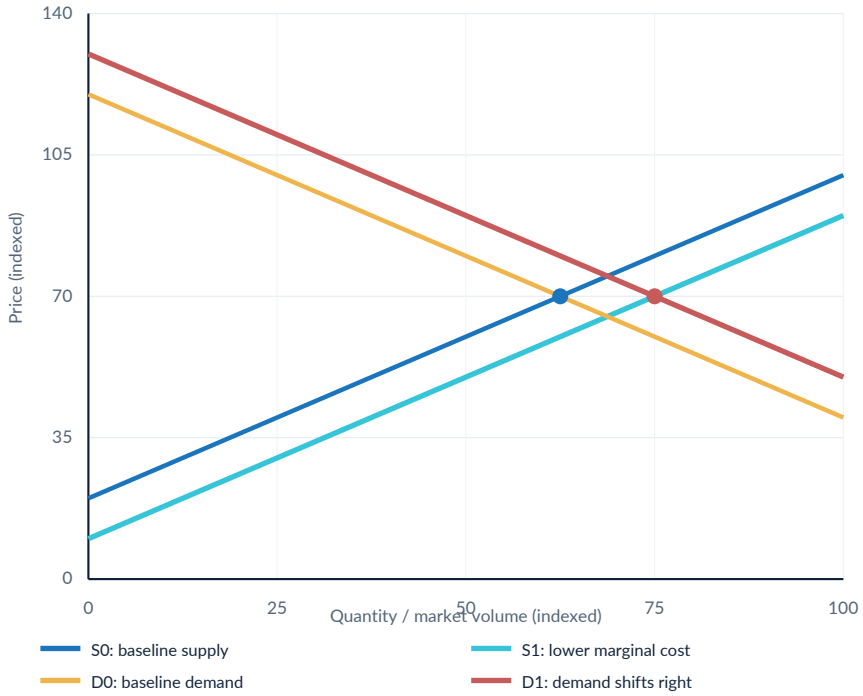
Prospectus boundary: This document proposes strategy and product architecture. It is not a securities prospectus, a commitment by Barclays, or evidence that a UK liquefaction project exists or is financeable.

ECONOMIC THESIS

Aggregate demand can rise as marginal cost falls

The investable question is whether demand growth absorbs incremental supply - not whether cost reductions mechanically dictate a lower market price.

Illustrative equilibrium model



THREE POSSIBLE OUTCOMES

- **Demand outruns supply**
  Demand shifts right more than the supply curve. Wholesale prices rise; sellers have pricing power.
- **Supply and demand balance**
  Lower cost expands volumes, while demand absorbs the extra gas. Prices may be broadly stable.
- **Supply outruns demand**
  Production and LNG capacity expand faster than consumption. Prices and producer margins can compress.

WHAT THE CURRENT EVIDENCE SAYS

Near term: the IEA Q3 2026 outlook expected global gas demand to decline around 0.5% in 2026 as high prices, fuel switching and disruption weigh on consumption. Global LNG supply was expected to be broadly flat for the year. [4]

Medium-term thesis: the IEA Gas 2025 outlook had projected average global gas demand growth of nearly 1.5% a year over 2024-2030. It is an earlier baseline, not the latest 2026 forecast; the divergence supports scenario-based underwriting. [5]

Cost-down underwriting must be project-specific. The UKCS is a mature basin, and aggregate costs can rise as fields deplete even where technology lowers the cost of a new project or individual operating task. Treat lower marginal cost as a hypothesis to validate, not a uniform property of all North Sea gas.

PRODUCT ARCHITECTURE

Launch with established benchmarks; add LNG structures in stages

A derivatives programme can begin before any new UK liquefaction capacity is built, provided contracts are labelled accurately and appropriately risk-managed.

Stage	Instrument / underlier	Target customer	Use case / guardrail
1 - Immediate	NBP futures, forwards and swaps Reference: UK NBP price curve. [3]	North Sea producers, utilities, industrial buyers	Fix a future selling or purchase price; match delivery periods and volumes to physical exposure.
2 - Expansion	NBP-TTF basis swaps Reference: British vs continental European gas prices.	Cross-border traders, utilities, portfolio holders	Hedge regional price divergence; test liquidity and benchmark basis before structuring.
2 - Expansion	TTF-JKM / LNG-linked spread structures	LNG portfolio holders, global gas traders	Manage Europe-Asia netback differences; do not assume cargoes are freely redirectable.
3 - Conditional	Proposed North Sea LNG forward or index swap	Future LNG project, offtakers, banks and traders	Only after physical liquefaction/export capacity, index methodology, liquidity, legal classification and governance are established.

RECOMMENDED INSTRUMENT SET

Fixed-price forwards

Revenue certainty for producers; purchase-cost certainty for buyers. Simplest first product.

Put options

Producer downside protection while preserving upside above the strike, net of premium and terms.

Collars

Lower-cost protection by exchanging some upside above a cap or other agreed level.

Spread swaps

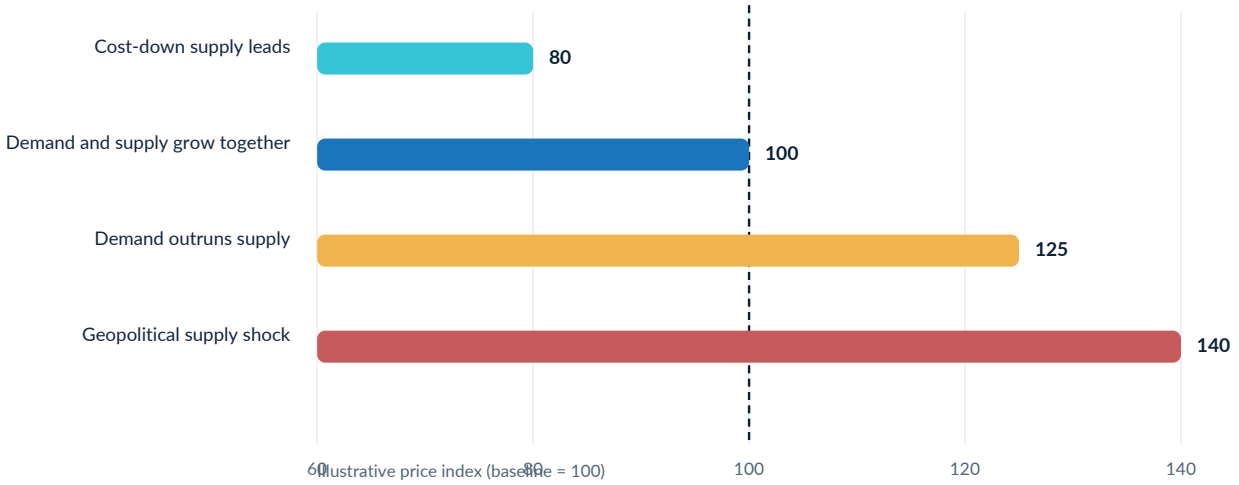
Hedge benchmark differentials where sales price and hedge reference do not match.

Index design principle: do not construct a bespoke “British LNG price” merely by relabelling NBP. LNG pricing must reflect liquefaction, fuel use, shipping, terminal access, delivery basis and quality. A new index needs an objective methodology, observable inputs, independent governance and sufficient market participation.

SCENARIO ANALYSIS

Price outcomes depend on the race between demand and supply

Use scenario ranges to size hedges and collateral needs; do not present these illustrative index levels as a forecast or as quoted market prices.



Scenario	Illustrative index	Interpretation	Potential product response
Cost-down supply expansion	80 (-20%)	New supply and efficiency outpace demand; benchmark price compresses.	Producer puts / partial forwards; buyers avoid over-hedging falling prices.
Demand and supply in balance	100 (flat)	Additional supply is absorbed; volumes rise, price remains near baseline.	Staggered forwards and collars; maintain flexibility.
Demand outruns supply	125 (+25%)	Industrial / power / LNG demand grows faster than deliverable supply.	Buyer call options or price caps; producers retain measured upside.
Geopolitical supply shock	140 (+40%)	Export, pipeline or shipping disruption raises scarcity and volatility.	Stress liquidity; options and robust counterparty limits.

How to read this table: The price index starts at 100 and the percentage shifts are deliberately illustrative. They are not actual NBP, TTF or JKM prices, probabilities, price targets or Barclays market views. A live desk model would use current forward curves, volatility surfaces, storage, cargo flows, FX and correlations.

Risk management should stress both price direction and price shape: prompt vs winter strips, NBP vs TTF, TTF vs JKM, basis widening, volatility spikes and changes in liquidity / margin requirements.

ILLUSTRATIVE ECONOMICS

Lower unit cost is valuable - until the benchmark price catches down

The commercial thesis depends on the difference between project-level cost improvement and market-wide price compression.

A. ILLUSTRATIVE PRODUCER CONTRIBUTION PER THERM

Case	Selling price	Illustrative delivered variable cost	Contribution per therm	Contribution on 1m therms
Baseline	60p	42p	18p	£180,000
Local cost-down; price unchanged	60p	35p	25p	£250,000
Market-wide supply growth; price falls	52p	35p	17p	£170,000

All prices and costs in this table are hypothetical teaching assumptions, not estimates of North Sea production cost, market quotations or a Barclays forecast. Contribution excludes fixed costs, tax, royalties, transport not included in the assumed cost, hedging and capital recovery.

B. WHAT A FIXED-PRICE HEDGE CHANGES

Future market price	Unhedged revenue on 1m therms	Illustrative locked-price revenue
40p / therm	£400,000	£600,000
60p / therm	£600,000	£600,000
80p / therm	£800,000	£600,000

Assumption: a producer locks an effective price of 60p per therm for 1 million therms, with perfectly matched volumes and timing and no fees, basis mismatch or default. This simplified fixed-price example demonstrates the trade-off: reduced downside is exchanged for surrendered upside. An option or collar would produce a different payoff.

For Barclays, fee and spread income should be modelled separately from commodity P&L.; The preferred risk profile is client-led intermediation, matched hedges and measured residual exposure - not a large unhedged directional bet on falling prices.

EXECUTION & CONTROLS

A staged launch protects the thesis from infrastructure and market risk

Proceed only when the product matches a verifiable physical exposure or a well-governed financial benchmark.

Risk	Why it matters	Required control
Physical feasibility	No domestic LNG chain currently; liquefaction/export capex and approvals can overwhelm a trading thesis.	Separate pre-feasibility workstream; no physical delivery promise before capacity is contracted and verified.
Demand reversal	2026 global demand forecast is weak; high prices can trigger fuel switching and industrial curtailment.	Use multiple demand cases; refresh macro assumptions and client demand weekly / monthly as appropriate.
Benchmark / basis risk	NBP, TTF and JKM can diverge as shipping, weather and terminal constraints change.	Hedge against the correct index; price basis risk and stress correlation breaks.
Volume / production risk	A producer may deliver less gas than the hedged amount.	Hedge only defensible volumes; caps, tolerances and production sensitivity limits.
Liquidity / collateral	Price spikes create margin calls even where long-term economics remain sound.	Stress liquidity, counterparty concentration and wrong-way risk; pre-agree collateral terms.
Regulatory / conduct / ESG	Structured derivatives and a new index raise conduct, market-abuse, disclosure and transition-risk issues.	Legal, compliance, benchmark-governance and climate-impact reviews before launch.

RECOMMENDED IMPLEMENTATION GATES

GATE 0

Physical and legal reality

Confirm the asset, production profile, delivery point, permits, transportation and - only if relevant - liquefaction / export chain.

GATE 1

NBP pilot

Design standardised producer and buyer hedges; validate client demand, execution, risk limits and clearing / collateral.

GATE 2

Cross-market products

Add basis derivatives when customer exposure and executable liquidity justify them.

GATE 3

New North Sea LNG index

Only proceed once physical LNG capacity or robust cash-settled reference methodology, independent governance and market depth exist.

REFERENCES & IMPORTANT NOTICE

Evidence base, assumptions and limitations

Source cut-off: 21 October 2026. Prices in scenario tables are illustrative unless explicitly attributed to a source.

[1] UK Parliament / Department for Energy Security and Net Zero

Written question HL16264, answered 27 April 2026. Confirms UK produces natural gas but does not produce LNG domestically; provides 2024 gross supply mix.

https://questions-statements.parliament.uk/written-questions/detail/2026-04-13/HL16264/

[2] UK Government / DESNZ / HM Treasury / NSTA

"Jackdaw gas field gets green light", published 21 October 2026. Official statement on production consent and expected peak domestic production uplift.

https://www.gov.uk/government/news/jackdaw-gas-field-gets-green-light

[3] ICE Futures Europe

UK NBP Natural Gas Futures contract specifications: NBP virtual trading point, physical delivery rights and monthly / quarterly / seasonal contract strips.

https://www.ice.com/products/910/UK-NBP-

[4] International Energy Agency

Gas Market Report, Q3-2026, executive summary, published 7 July 2026. Near-term global demand and LNG supply outlook; regional price and flow dynamics.

https://www.iea.org/reports/gas-market-report-q3-2026/executive-summary

[5] International Energy Agency

Gas 2025, executive summary. Earlier medium-term baseline projected nearly 1.5% average annual gas demand growth over 2024-2030; used here as a historical baseline, not the latest 2026 forecast.

https://www.iea.org/reports/gas-2025/executive-summary

[6] North Sea Transition Authority

Production and expenditure projections, February 2026 dataset and related UKCS outlook material. Relevant to field-level production, expenditure and basin decline assumptions.

https://www.nstauthority.co.uk/data-and-insights/insights-and-analysis/production-and-expenditure-projections/

IMPORTANT NOTICE

This document has been prepared as a strategic discussion draft only. It is not an official Barclays publication, an invitation or offer to invest, a securities prospectus, a recommendation to enter into any transaction, or legal, tax, accounting or investment advice. No transaction is being offered by this document. It has not been approved by a regulator. Any actual derivatives activity would be subject to suitability and appropriateness requirements where applicable, applicable derivatives, benchmark, market-abuse, clearing, collateral, capital and conduct rules, and the policies of the relevant institutions. The document contains forward-looking statements, scenarios and illustrative calculations. Actual market conditions, gas demand, field output, marginal costs, benchmark relationships, infrastructure costs, approvals and price outcomes may differ materially. References to "new North Sea LNG" describe a conditional future concept; they do not assert that UK liquefaction or export capacity currently exists. All model assumptions should be independently verified before investment, financing or trading decisions.

End of document | Prepared for strategic discussion

END